f is continuous at x_0 ① $f(x_0)$ exists. ② $\lim_{x\to x_0} f(x)$ exists.
③ $\lim_{x\to x_0} f(x) = f(x_0)$

Continuity:

$\forall \varepsilon>0,\ \delta>0$ such that $|x-x_0|<\delta \Rightarrow |f(x)-f(x_0)|<\varepsilon$.

Uniform continuity: $f: S\to\mathbb{R}$ be a function

$\forall \varepsilon>0,\ \delta>0$, such that $x_1, x_2 \in S$, and $|x_1-x_2|<\delta$

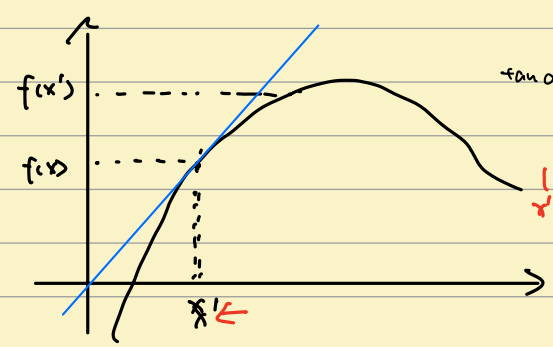
$\Rightarrow |f(x_1)-f(x_2)|<\varepsilon$.

$f(x)=\frac{1}{x}$ $(0,1)$

切线

tangent line

Differentiation



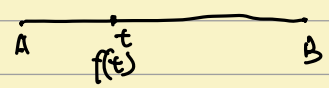
$\tan\alpha = \frac{f(x)-f(x')}{x-x'}$

$\Downarrow$

$\lim_{x'\to x} \frac{f(x)-f(x')}{x-x'}$

$f(t)$

速度 $= \frac{\text{距离}}{\text{时间}}$



$\frac{f(t)-f(0)}{t-0}$ $\lim_{\Delta t\to 0} \frac{f(t+\Delta t)-f(t)}{\Delta t}$

Defn: A function $f:(a,b)\to\mathbb{R}$ is differentiable at a point $x_0\in(a,b)$ if the limit $\lim_{h\to 0}\frac{f(x_0+h)-f(x_0)}{h}$ exist. The limit is called the derivative of f, and denoted $f'(x_0)$.

$\Delta x = x - x_0 \qquad \Delta y = f(x) - f(x_0)$

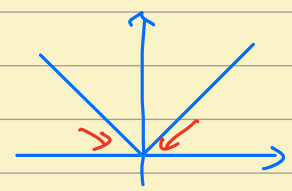
$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

E.g. $f(x) = x^2 \qquad x = 1$

$$f'(1) = \lim_{\Delta x \to 0} \frac{f(1+\Delta x) - f(1)}{\Delta x} = \lim_{\Delta x \to 0} \frac{(1+\Delta x)^2 - 1}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{2\Delta x + \Delta x^2 + 1 - 1}{\Delta x} = \lim_{\Delta x \to 0} (2 + \Delta x) = 2$$

$f(x) = |x| \qquad x = 0$



$$f'(0) = \lim_{\Delta x \to 0^+} \frac{|0 + \Delta x| - |0|}{\Delta x} = 1$$

$$f'(0) = \lim_{\Delta x \to 0^-} \frac{|0 + \Delta x| - |0|}{\Delta x} = \lim_{\Delta x \to 0^-} \frac{-\Delta x}{\Delta x} = -1$$

By the definition of derivative, the limit $f'(0)$ does not exist.

Thm: If a function is differentiable at a point x_0, then it is continuous at x_0.

Proof: Since f is differentiable at x_0, then

$$f'(x) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

To prove $\lim_{x \to x_0} f(x) = f(x_0) \iff \lim_{h \to 0} f(x_0 + h) = f(x_0)$

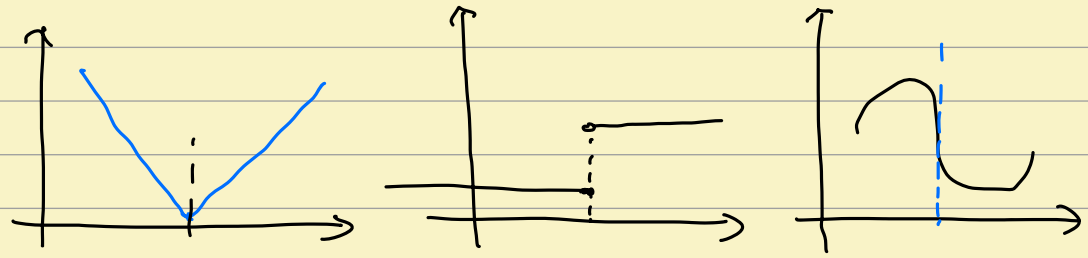
we have $\lim_{h \to 0} \left(f(x_0 + h) - f(x_0) \right) = \lim_{h \to 0} \left(\frac{f(x_0 + h) - f(x_0)}{h} \cdot h \right)$

$$= \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h} \lim_{h \to 0} h$$

$$= f'(x_0) \cdot 0 = 0$$

$$\Rightarrow \lim_{h \to 0} f(x_0 + h) = f(x_0) \Rightarrow \lim_{x \to x_0} f(x) = f(x_0)$$

Corollary : If f is not continuous at x_0, then it is not differentiable at x_0.



corner discontinuity vertical tangent

$\Delta x = x - x_0 \quad \Delta y = f(x) - f(x_0)$

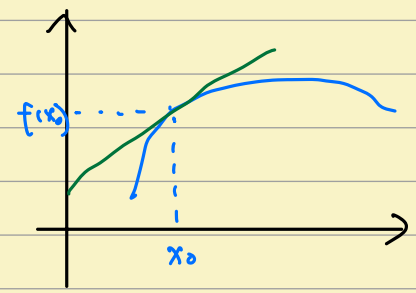
$$f'(x_0) = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

$$\varepsilon = \frac{\Delta y}{\Delta x} - f'(x) \qquad \varepsilon \to 0 \text{ as } \Delta x \to 0$$

$$\Delta y = f'(x_0)\Delta x + \varepsilon$$

$$f(x) - f(x_0) = f'(x_0)(x - x_0) + \varepsilon$$

$$\Rightarrow \quad f(x) = f(x_0) + f'(x_0)(x - x_0) + \varepsilon$$



Differentiation rules:

$f(x) = c$ is a constant function, then $f'(x) = 0 \quad \forall x \in \mathrm{dom}(f)$

Proof: $$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} \frac{c - c}{h} = 0$$

E.g. $(x^n)' = n x^{n-1}$ $\qquad (\log_a x)' = \frac{1}{x} \log_a e$

$(\sin x)' = \cos x$

$(\cos x)' = -\sin x$

① $(x^n)' = nx^{n-1}$ $\quad \dfrac{\Delta y}{\Delta x} = \dfrac{(x_0+\Delta x)^n - x_0^n}{\Delta x_0} = \dfrac{x^n - x_0^n}{x - x_0}$

$$x^n - x_0^n = (x - x_0)\left(x^{n-1} + x^{n-2}x_0 + x^{n-3}x_0^2 + \cdots x x_0^{n-2} + x_0^{n-1}\right)$$

$$= x^{n-1} + x^{n-2}x_0 + x^{n-3}x_0^2 + \cdots x x_0^{n-2} + x_0^{n-1}$$

$$\lim_{x \to x_0}\left(x^{n-1} + x^{n-2}x_0 + x^{n-3}x_0^2 + \cdots x x_0^{n-2} + x_0^{n-1}\right)$$

$$= x_0^{n-1} + \underbrace{x_0^{n-2}\cdot x_0}_{x_0^{n-1}} \cdots = n x_0^{n-1}$$

$$(x_0^n)' = n x_0^{n-1}$$

② $(\sin x)' = \cos x$

$$\frac{\sin(x+\Delta x) - \sin x}{\Delta x} = \frac{\sin x \cos\Delta x + \sin\Delta x \cos x - \sin x}{\Delta x}$$

$$= \frac{\sin\Delta x}{\Delta x}\cos x + \sin x \frac{\cos\Delta x - 1}{\Delta x}$$

$$= \frac{\sin\Delta x}{\Delta x}\cos x + \sin x \frac{\cos^2\Delta x - 1}{(\cos\Delta x + 1)\Delta x}$$

$$\lim_{\Delta x \to 0}\frac{\sin\Delta x}{\Delta x}\cos x = \cos x \qquad \lim_{\Delta x \to 0}\sin x \frac{\cos^2\Delta x - 1}{\Delta x}\frac{1}{\cos\Delta x + 1}$$

$$= \sin x \lim_{\Delta x \to 0}\frac{\cos^2\Delta x - 1}{\Delta x}\frac{1}{\cos\Delta x + 1}$$

$$= \sin x \lim_{\Delta x \to 0}\frac{\cos^2\Delta x - 1}{\Delta x}\lim_{\Delta x \to 0}\frac{1}{\cos\Delta x + 1} = 0$$

$$\lim_{\Delta x \to 0}\frac{\sin(x+\Delta x) - \sin x}{\Delta x} = \cos x$$

$(\cos x)' = -\sin x$

$$\lim_{\Delta x \to 0}\frac{\cos(x+\Delta x) - \cos x}{\Delta x} = \lim_{\Delta x \to 0}\frac{\cos x \cos\Delta x - \sin x \sin\Delta x - \cos x}{\Delta x}$$

$$= \lim_{\Delta x \to 0} -\sin x \frac{\sin\Delta x}{\Delta x} + \cos x \frac{\cos\Delta x - 1}{\Delta x}$$

$$= -\sin x$$

$$(\log_a x)' = \frac{1}{x}\log_a e$$

$$\frac{\log_a(x+\Delta x) - \log_a x}{\Delta x} = \frac{1}{\Delta x}\log_a \frac{x+\Delta x}{x} = \frac{1}{\Delta x}\log_a\left(1+\frac{\Delta x}{x}\right)$$

$$= \frac{x}{\Delta x}\cdot\frac{1}{x}\log_a\left(1+\frac{\Delta x}{x}\right)$$

$$= \frac{1}{x}\log_a\left(1+\frac{\Delta x}{x}\right)^{\frac{x}{\Delta x}}$$

$$\lim_{\Delta x\to 0}\frac{1}{x}\log_a\left(1+\frac{\Delta x}{x}\right)^{\frac{x}{\Delta x}} = \frac{1}{x}\log_a\left(\lim_{\Delta x\to 0}\left(1+\frac{\Delta x}{x}\right)^{\frac{x}{\Delta x}}\right)$$

$$= \frac{1}{x}\log_a e \qquad a = e$$

$$(\ln x)' = \frac{1}{x}\cancel{\log_e e}^{\,1} = \frac{1}{x}$$

$$(e^x)' = e^x \qquad \lim_{\Delta x\to 0}\frac{e^{x+\Delta x} - e^x}{\Delta x} = e^x\lim_{\Delta x\to 0}\frac{(e^{\Delta x}-1)}{\Delta x} = e^x$$

The algebra of differentiation

① $(f(x)+g(x))' = f'(x) + g'(x)$

$$\lim_{\Delta x\to 0}\frac{f(x+\Delta x) + g(x+\Delta x) - f(x) - g(x)}{\Delta x}$$

$$= \lim_{\Delta x\to 0}\frac{f(x+\Delta x) - f(x)}{\Delta x} + \lim_{\Delta x\to 0}\frac{g(x+\Delta x) - g(x)}{\Delta x}$$

② $(c\cdot f(x))' = c\cdot f'(x)$

$$\lim_{\Delta x\to 0}\frac{c f(x+\Delta x) - c f(x)}{\Delta x} = c\lim_{\Delta x\to 0}\frac{f(x+\Delta x) - f(x)}{\Delta x}$$

③ $(fg)'(x) = f'(x)\cdot g(x) + f(x)\cdot g'(x)$

$$\lim_{\Delta x\to 0} \frac{f(x+\Delta x)\cdot g(x+\Delta x) - f(x)\cdot g(x)}{\Delta x}$$

$$= \lim_{\Delta x\to 0} \frac{1}{\Delta x}\Big(f(x+\Delta x)\cdot g(x+\Delta x) - f(x+\Delta x)g(x) + f(x+\Delta x)g(x) - f(x)\cdot g(x)\Big)$$

$$= \lim_{\Delta x\to 0} \frac{1}{\Delta x} f(x+\Delta x)\big(g(x+\Delta x) - g(x)\big) + \lim_{\Delta x\to 0} g(x)\frac{1}{\Delta x}\big(f(x+\Delta x) - f(x)\big)$$

$$= \lim_{\Delta x\to 0} f(x+\Delta x)\cdot \lim_{\Delta x\to 0}\frac{g(x+\Delta x)-g(x)}{\Delta x} + g(x)\lim_{\Delta x\to 0}\frac{f(x+\Delta x)-f(x)}{\Delta x}$$

$$= f(x)\,g'(x) + g(x)\cdot f'(x)$$

① $\left(\frac{f}{g}\right)'(x) = \dfrac{f'(x)\cdot g(x) - f(x)\cdot g'(x)}{[g(x)]^2}$ $\quad g(x) \neq 0$

$$\lim_{\Delta x\to 0} \frac{\frac{f(x+\Delta x)}{g(x+\Delta x)} - \frac{f(x)}{g(x)}}{\Delta x} = \lim_{\Delta x\to 0}\frac{f(x+\Delta x)\cdot g(x) - f(x)\cdot g(x+\Delta x)}{\Delta x\, g(x+\Delta x)\, g(x)}$$

$$= \lim_{\Delta x\to 0}\frac{f(x+\Delta x)\cdot g(x) - f(x)\cdot g(x+\Delta x)}{\Delta x\, g(x+\Delta x)\, g(x)}$$

$$\lim_{\Delta x\to 0}\frac{1}{g(x+\Delta x)\,g(x)}\lim_{\Delta x\to 0}\frac{1}{\Delta x}\Big(f(x+\Delta x)\cdot g(x) - f(x)\cdot g(x+\Delta x)\Big)$$

$$= \frac{1}{[g(x)]^2}\cdot\lim_{\Delta x\to 0}\frac{1}{\Delta x}\Big(f(x+\Delta x)\cdot g(x) - g(x)f(x) + f(x)g(x) - f(x)\cdot g(x+\Delta x)\Big)$$

$$= \frac{1}{[g(x)]^2}\lim_{\Delta x\to 0} g(x)\frac{f(x+\Delta x) - f(x)}{\Delta x} + \lim_{\Delta x\to 0} f(x)\frac{g(x) - g(x+\Delta x)}{\Delta x}$$

$$= \frac{1}{[g(x)]^2}\big(f'(x)g(x) - f(x)g'(x)\big)$$

$\bar{D}x$. $(\tan x)' = \sec^2 x$ $\qquad$ $(\cot x)' = -\csc^2 x$

$(\sec x)' = \sec x \tan x$ $\qquad$ $(\csc x)' = -\csc x \cot x$

hint: $\tan x = \dfrac{\sin x}{\cos x}$, $\cot x = \dfrac{\cos x}{\sin x}$, $\sec x = \dfrac{1}{\cos x}$, $\csc x = \dfrac{1}{\sin x}$

$(\tan x)' = \sec^2 x$

$$\lim_{\Delta x \to 0} \frac{\tan(x+\Delta x) - \tan x}{\Delta x} = \lim_{\Delta x \to 0} \frac{\dfrac{\sin(x+\Delta x)}{\cos(x+\Delta x)} - \dfrac{\sin x}{\cos x}}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{1}{\Delta x} \frac{\sin(x+\Delta x)\cos x - \cos(x+\Delta x)\cdot \sin x}{\cos(x+\Delta x)\cdot \cos x}$$

$$= \lim_{\Delta x \to 0} \frac{1}{\Delta x} \sin(x+\Delta x - x) \cdot \lim_{\Delta x \to 0} \frac{1}{\cos(x+\Delta x)\cdot \cos x}$$

$$= 1 \cdot \frac{1}{\cos^2 x} = \sec^2 x$$

$(\cot x)' = -\csc^2 x$

$$\lim_{\Delta x \to 0} \frac{\cot(x+\Delta x) - \cot x}{\Delta x} = \lim_{\Delta x \to 0} \frac{\dfrac{\cos(x+\Delta x)}{\sin(x+\Delta x)} - \dfrac{\cos x}{\sin x}}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{1}{\Delta x} \frac{\sin x \cos(x+\Delta x) - \cos x \sin(x+\Delta x)}{\sin(x+\Delta x)\cdot \sin x}$$

$$= \lim_{\Delta x \to 0} \frac{\sin(x-(x+\Delta x))}{\Delta x} \cdot \lim_{\Delta x \to 0} \frac{1}{\sin(x+\Delta x)\cdot \sin x} = -\csc^2 x$$

$(\sec x)' = \sec x \tan x$

$$\lim_{\Delta x \to 0} \frac{\sec(x+\Delta x) - \sec x}{\Delta x} = \lim_{\Delta x \to 0} \frac{\dfrac{1}{\cos(x+\Delta x)} - \dfrac{1}{\cos x}}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{1}{\Delta x} \frac{\cos x - \cos(x+\Delta x)}{\cos(x+\Delta x)\cos x} = \lim_{\Delta x \to 0} \frac{1}{\Delta x} \frac{\cos x - \cos x \cos\Delta x + \sin x \sin\Delta x}{\cos(x+\Delta x)\cos x}$$

$$= \lim_{\Delta x \to 0} \frac{1}{\Delta x} \frac{\cancel{\cos x}(1-\cos\Delta x)}{\cos x \cos(x+\Delta x)} + \lim_{\Delta x \to 0} \frac{1}{\Delta x} \frac{\sin x \sin\Delta x}{\cos(x+\Delta x)\cos x}$$

$$= 0 + \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x} = \tan x \sec x$$

$$= \lim_{x\to 0} \frac{1-\cos x}{x} = \lim_{x\to 0} \frac{1-\cos^2 x}{x(1+\cos x)} = \lim_{x\to 0} \cancel{\frac{\sin x}{x}}^{\to 1} \cdot \cancel{\frac{\sin x}{(1+\cos x)}}^{\to 0}$$

$$(\csc x)' = -\csc x \cot x \qquad \csc x = \frac{1}{\sin x}$$

$$\lim_{\Delta x\to 0} \frac{\csc(x+\Delta x) - \csc x}{\Delta x} = \lim_{\Delta x\to 0} \frac{\frac{1}{\sin(x+\Delta x)} - \frac{1}{\sin x}}{\Delta x}$$

$$= \lim_{\Delta x\to 0} \frac{1}{\Delta x} \frac{\sin x - \sin(x+\Delta x)}{\sin x \sin(x+\Delta x)} = \lim_{\Delta x\to 0} \frac{1}{\Delta x} \frac{\sin x - \sin x\cos\Delta x - \cos x \sin\Delta x}{\sin x \sin(x+\Delta x)}$$

$$= \lim_{\Delta x\to 0} \frac{1}{\Delta x} \frac{\cancel{\sin x}(1-\cos\Delta x)}{\cancel{\sin x}\sin(x+\Delta x)} - \lim_{\Delta x\to 0} \frac{1}{\Delta x} \frac{\cos x \sin\Delta x}{\sin x \sin(x+\Delta x)}$$

$$= 0 - \frac{\cos x}{\sin x} \frac{1}{\sin x} = -\cot x \cdot \csc x$$

$$\lim_{\Delta x\to 0} \frac{1-\cos\Delta x}{\Delta x} = \frac{1-\cos^2\Delta x}{\Delta x(1+\cos\Delta x)} = \frac{\sin^2\Delta x}{\Delta x} \frac{1}{1+\cos\Delta x} = 0$$